

Software Development And Documentation Project

Smart Transportation System Documentation

Members

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Course

Software Development And Documentation

Instructor

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Section

Section 1

Semester

Second Semester 2025/2026

1. User Stories

1.1 Student

- As a student, I want to log in so that I can access the application.
- As a student, I want to reset my password in case I forget it.
- As a student, I want to open the dashboard and choose between going to MEU or leaving it.
- As a student, I want to check the bus timetables so I know when the buses leave.
- As a student, I want to see the bus routes on the map so that I can understand the bus's destination.
- As a student, I want to track the bus route in real time so I know when it's approaching me.
- As a student, I would like to update my profile information as needed.
- As a student, I want an easy way to communicate with the driver or support team.
- As a student, I would like to evaluate the service based on my experience.

1.2 Driver

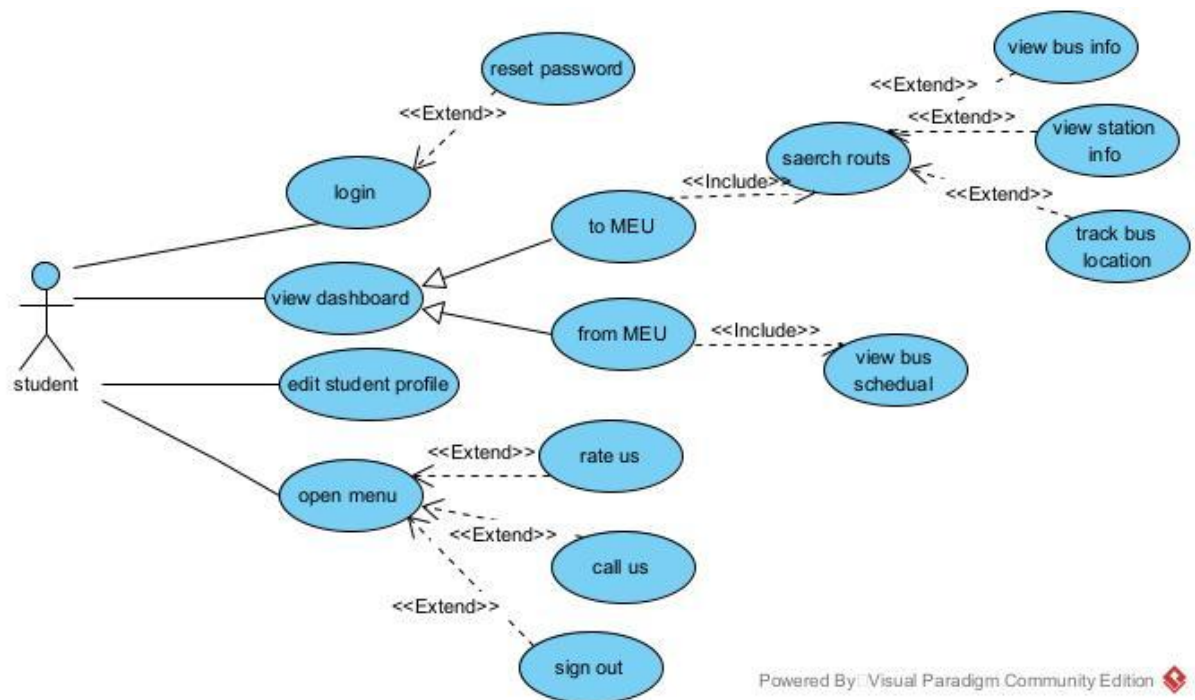
- As a driver, I want to log in so I can see my tasks.
- As a driver, I want to reset my password if I can't access my account.
- As a driver, I would like to display my dashboard to manage my business.
- As a driver, I want to change my status (active or inactive) based on my availability.
- As a driver, I want to review my assigned trips to know what to do next.
- As a driver, I want to easily start a trip from the app.
- As a driver, I want to use the navigation system during the trip so I don't miss any stops.
- As a driver, I want to update my personal information if there are any changes.
- As a driver, I want to report any problems during the trip quickly.

1.3 Admin

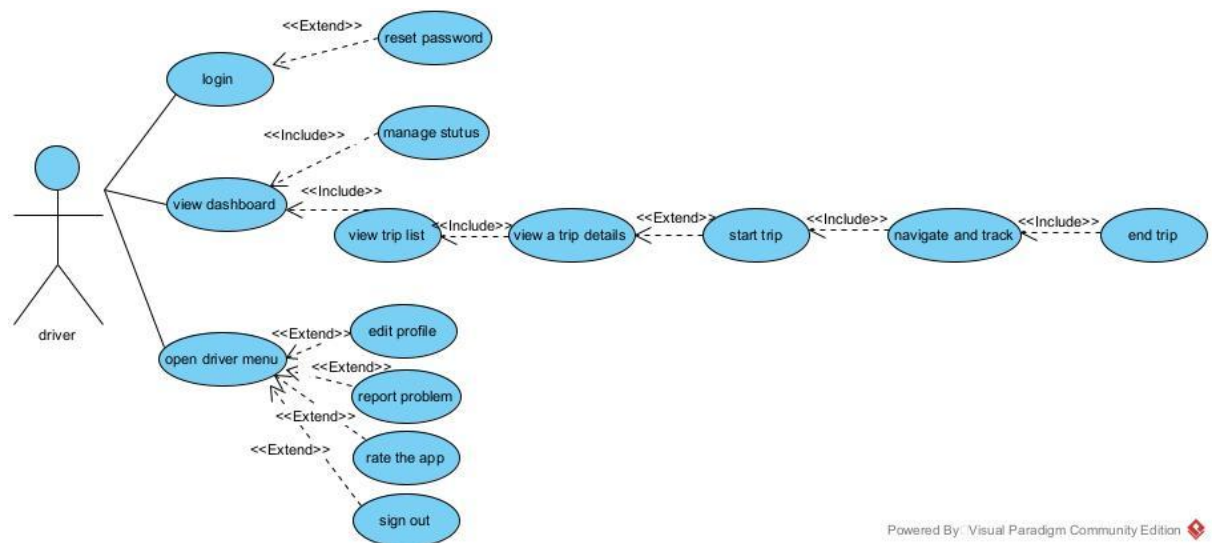
- As an administrator, I want to log in to control the system.
- As an administrator, I want to log out securely after I finish my work.
- As an administrator, I want to see all buses in one place.
- As an administrator, I want to edit bus data as needed.
- As an administrator, I want to view driver information to manage them properly.
- As an administrator, I want to assign trips and tasks to drivers.
- As an administrator, I want to easily create and update schedules.
- As an administrator, I want to track buses in real time to see what's happening.
- As an administrator, I want to receive alerts in case of any problems.
- As an administrator, I want to generate reports to evaluate the system.
- As an administrator, I want to update my profile and settings.

2. Use Cases

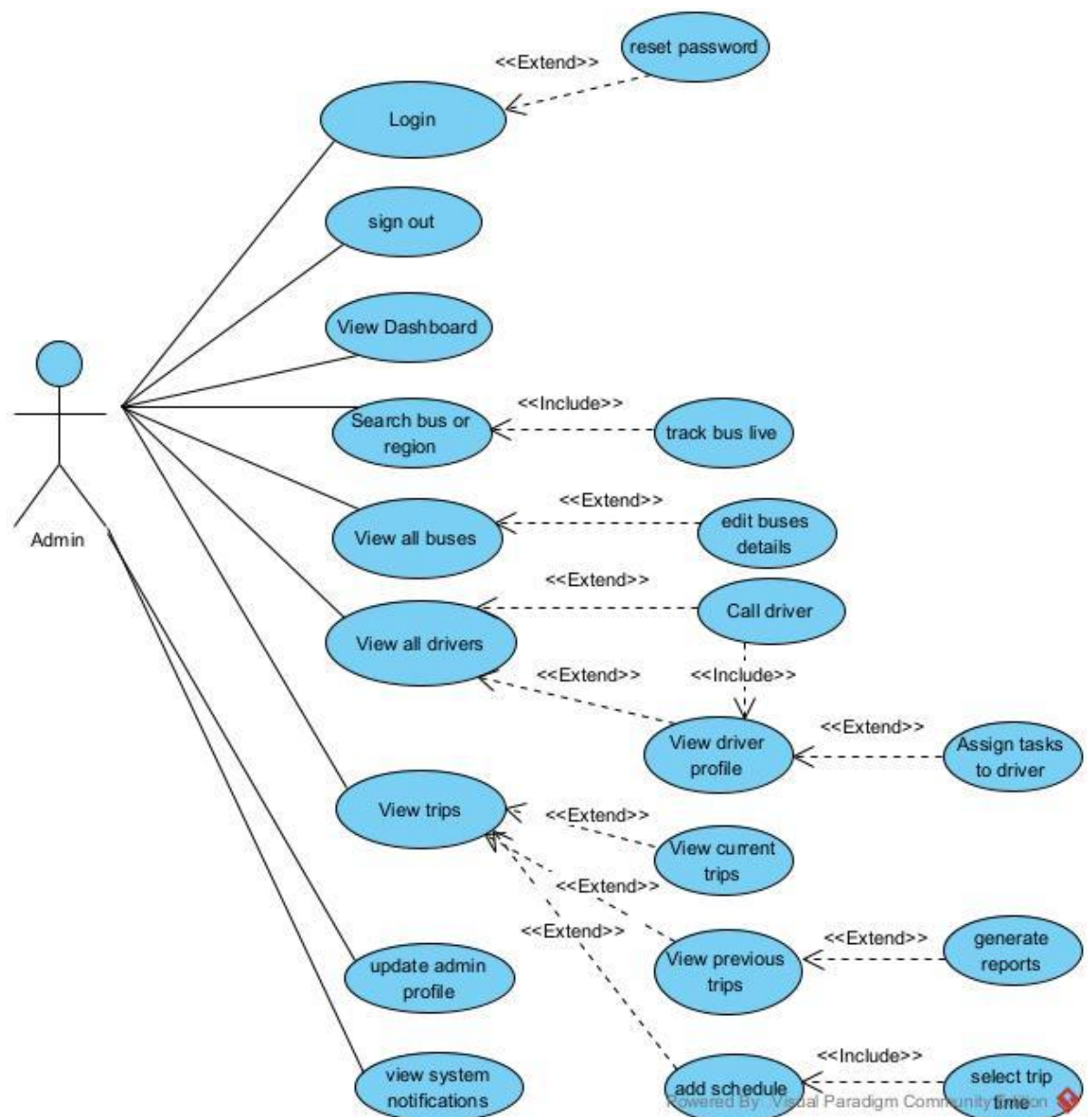
2.1 Student Use Case



2.2 Driver Use Case



2.3 Admin Use Case



3.4 Use Case Description

1. log in

Use Case ID	UC001
Use Case Name	log in
Created By	Sarah Shahrour
Date Created	25/4/2026
Brief Description	Allows the student to securely access the application using their username and password.

Primary Actor	Student
Secondary Actor	-
Pre-condition	The user has a valid account (username and password).
Post-Condition	The student is redirected to the dashboard.
Main Flows	1. The student opens the app. 2. The login screen appears. 3. The student enters their username and password. 4. The student clicks the “Login” button. 5. The system verifies the credentials. 6. If correct, the system shows the dashboard.
Alternative Flow	3a. If credentials are invalid: • System displays error message: “Invalid username or password”. • Returns to step 2.

2. View Buses Schedule

Use Case ID	UC002
Use Case Name	View Buses Schedule
Created By	Noor shananeer
Date Created	25/4/2026
Brief Description	Displays a full schedule of buses leaving MEU.
Primary Actor	Student
Secondary Actor	-
Pre-condition	“From MEU” option is selected.
Post-Condition	The bus schedule is displayed.
Main Flows	1. The student clicks on the schedule option. 2. The full list of routes, areas, and times is shown.

3. Add Schedule

Use Case ID	UC003
Use Case Name	Add Schedule
Created By	Sarah Shahrour
Date Created	25/4/2026
Brief Description	Allows the administrator to add a new trip schedule including trip time, destination, and bus number.
Primary Actor	Admin
Secondary Actor	-
Pre-condition	• The administrator must be logged in. • Bus and destination data must be available.
Post-Condition	The schedule is saved and added to the list of trips.
Main Flows	1. The administrator opens the “Add Schedule” page. 2. Enters trip details: • Destination • Trip time • Bus number 3. Click the “Add” button. 4. The system saves the schedule and confirms success.
Alternative Flow	2a. Incomplete fields: a. The system displays a warning for missing information. b. The administrator is prompted to complete the required fields.

4. View All Buses

Use Case ID	UC004
Use Case Name	View All Buses
Created By	Noor Shananeer
Date Created	25/4/2026
Brief Description	Displays a list of all buses with relevant details.
Primary Actor	Admin
Secondary Actor	-
Pre-condition	Admin must be logged in.

Post-Condition	The bus list is shown on the table.
Main Flows	1. Admin selects “Buses” from the menu. 2. The system shows a table with bus details: number, route, time, capacity, notes, etc.

5.view trip list

Use Case ID	UC005
Use Case Name	View trip list
Created By	Sarah Shahrour
Date Created	25/4/2026
Brief Description	Allows the driver to view the list of all assigned trips.
Primary Actor	Driver
Secondary Actor	-
Pre-condition	The driver must be logged in and on the dashboard screen.
Post-Condition	A list of all upcoming or active trips is displayed.
Main Flows	1. Driver navigates to the trip list section from the dashboard. 2. System fetches all assigned trips for the driver. 3. Trips are displayed with summary info (date, route, status). 4. Driver may select a trip to view its details.
Alternative Flow	2a. If there are no assigned trips: • System shows a message: “No trips assigned.”

6. Start Trip

Use Case ID	UC006
Use Case Name	Start Trip
Created By	Noor Shananeer
Date Created	25/4/2026
Brief Description	Begin the selected trip and activate trip tracking.
Primary Actor	Driver

Secondary Actor	-
Pre-condition	A trip must be selected.
Post-Condition	Trip status is set to "In Progress".
Main Flows	1. Driver presses "Start Trip". 2. System marks trip as started. 3. Navigation and tracking begin.

3. Functional Specification

1. Actors

- Student
- Driver
- Administrator

2. Student Functional Specifications

2.1 Authentication

FR1 – Student Login

Description: Allow students to log in using username and password

Inputs: Username, Password

Process: The system validates the entered credentials

Output: Access granted or error message

FR2 – Forgot Password

Description: Allow students to reset password via university email

Inputs: University email

Process: The system sends a verification code, validates it, then allows password reset

Output: Password reset confirmation

2.2 Dashboard

FR3 – View Dashboard

Description: Display dashboard after successful login

Output: Two options are shown (To MEU, From MEU)

2.3 Transportation Features

FR4 – View Bus Schedule (From MEU)

Description: Display bus schedules provided by the administrator

Output: Bus number and schedule details

FR5 – Interactive Map (To MEU)

Description: Display an interactive map with routes and stations

Features: Route selection, station locations, bus details

2.4 Profile and Menu

FR6 – Manage Profile

Description: Allow students to edit personal information

Editable fields: Student number, major, home location, profile picture

Menu options: Rate Us, Call Us, Sign Out

3. Driver Functional Specifications

3.1 Authentication

FR7 – Driver Login

Description: Allow drivers to log in using valid credentials

FR8 – Driver Forgot Password

Description: Allow drivers to reset password via personal email

3.2 Dashboard and Status

FR9 – Driver Dashboard

Description: Display dashboard and allow driver to change status

Options: Active, Inactive

3.3 Trip Management

FR10 – View Assigned Trips

Description: Display assigned trips

Action: Driver can start a trip using “Begin Trip”

FR11 – Trip Tracking (To MEU)

Description: Provide real-time tracking during the trip

Details: Navigation, next station, passenger list

FR13 – Trip Tracking (From MEU)

Description: Track trip until all stations are completed

3.4 Profile and Support

FR12 – Driver Profile

Description: Allow drivers to edit personal information and report issues

4. Administrator Functional Specifications

4.1 Authentication

FR14 – Admin Login and Password Recovery

Description: Allow administrators to log in and reset password

FR15 – Logout

Description: Allow administrators to log out securely

4.2 Dashboard and Monitoring

FR16 – System Overview

Description: Display real-time system data

Details: Bus locations, routes, stations

FR17 – Interactive Map

Description: Display live map with bus tracking

Features: Zoom and pan

FR18 – Search

Description: Allow searching for buses or regions

4.3 Bus Management

FR19 – View Bus Data

Description: Display all bus details

Details: Bus number, destination, capacity, passenger count

FR20 – Edit Bus Details

Description: Allow administrators to update bus information

4.4 Driver Management

FR22 – View Drivers

Description: Display list of drivers

FR23 – Driver Profile

Description: Display detailed driver information

FR24 – Call Driver

Description: Allow administrator to call drivers

FR25 – Assign Tasks

Description: Assign and modify driver tasks

4.5 Trip Management

FR26 – View Ongoing Trips

Description: Display current trips with full details

FR27 – View Past Trips

Description: Display historical trip records

FR28 – Add Trip Schedule

Description: Allow administrators to create new schedules

4.6 Profile and Settings

FR29 – Admin Profile Management

Description: Allow administrators to edit profile information

FR30 – System Settings

Description: Configure system settings and notifications

4.7 Monitoring and Alerts

FR31 – Real-Time Updates

Description: Display live operational data

FR32 – Alerts

Description: Notify administrators of critical issues

Examples: Fuel issues, delays, violations

4.8 Reporting

FR33 – Reports and Analytics

Description: Generate performance and historical reports

4.9 Security

FR34 – Security Controls

Description: Apply security measures

Includes: Encryption, password policies, audit logs